Moon Evolution Odyssey

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Analysing habitability potential on icy moons Europa and Enceladus by studying their geophysical characteristics and tidal heating

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Mentees

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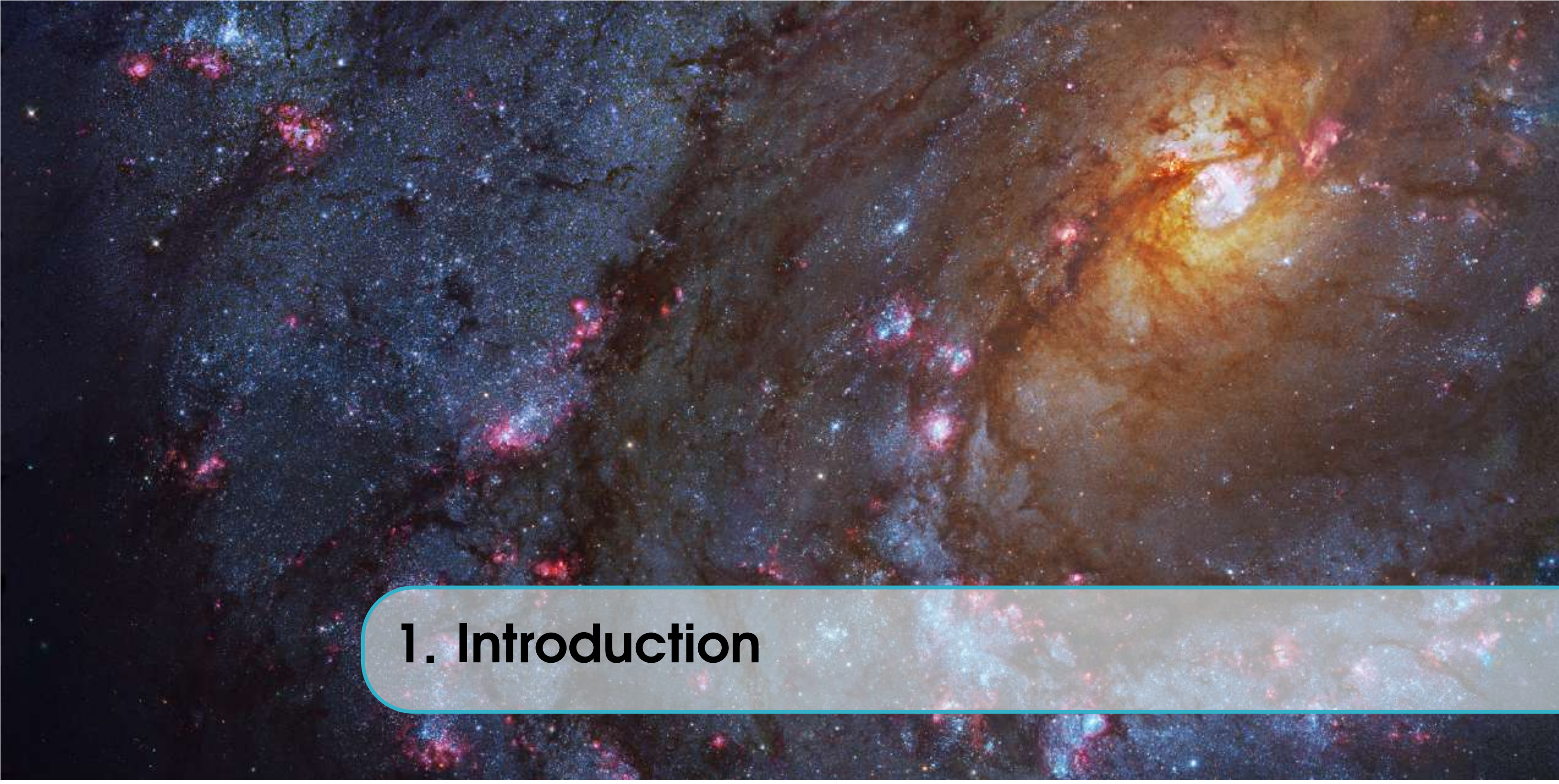


1. Introduction

2.

You can find some example test I tested with Matlab and with Python in this web page: `https://github.com/LaurethTeX/Clustering/blob/master/Methods.md`, also there is a huge amount of information on the internet about this but here are two pages you might find useful:

- Superpixel: Empirical Studies and Applications
  `http://ttic.uchicago.edu/~xren/research/superpixel/`
- Segmentation Algorithms in scikits-image
  `http://peekaboo-vision.blogspot.ca/2012/09/segmentation-algorithms-in-scikits-image.html`

Also there is one article (from IEEE) I found about and might interest you, it's pure computer science,

- Normalized Cuts and Image Segmentation
  `http://www.cs.berkeley.edu/~malik/papers/SM-ncut.pdf`

2.0.1 PCA

Welcome to Astronomy where you will find more acronyms than words to mention something on articles, lots of fun!, well in this case PCA stands for Principal Component Analysis, the objective of this method is to reduce dimensionality, transform the data to another space where is can be manipulated and reduced, there are multiple examples of work that has been done in astronomy applying this technique.

Therefore, the idea of applying this method is that if we have multiple-wavelength images of the same target and transform them to PCA space then we will have less dimensionality and it will be easier to process all the data and fins valuable information.[1]

An example article, where they explain how to apply PCA on multi-wavelength images and also mentions the pros and cons of using it.

[1] Before I forget to mention, later I discovered that PCA is not commonly used for data mining preprocessing because it is hard to interpret the information in the output result. Imagine clusters of data on PCA space, how do you make sense to that?

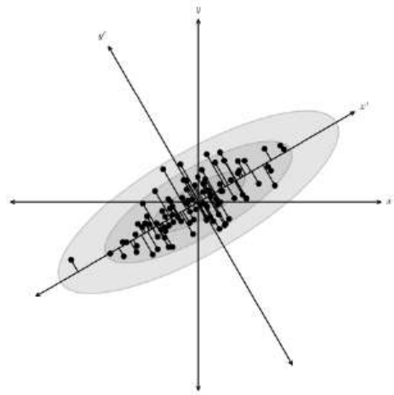


Figure 2.1: A distribution of points drawn from a bivariate Gaussian and centred on the origin of x and y. PCA defines a rotation such that the new axes (x' and y') are aligned along the directions of maximal variance (the principal components) with zero covariance. This is equivalent to minimizing the square of the perpendicular distances between the points and the principal components

- Preserving Structure in Multi-wavelength Images of Extended Objects
  `http://arxiv.org/abs/1101.1679v1`

There's a whole section that talks about this subject with a machine learning approach as a preprocessing step in this nice book,

- Ivezić, Ž. and Connolly, A.J. and Vanderplas, J.T. and Gray, A., *Statistics, Data Mining and Machine Learning in Astronomy*, Princeton University Press, Princeton, NJ, 2014.

2.1 Hypothesis

Our data looks like the images on Fig.2.2, and it contains data from let's say a determined galaxy at different wavelengths, if we assume that the galaxy contains various regions that relate to interstellar objects that can tell, how stars are formed, where, how stars die, where was a star, and other mysteries, I guess we can assume that those certain regions can be identified because they share similar characteristics, the ideas is to find how a galaxy is made from, its contents, apply the concept of the superpixel idea in 3D superpixels.

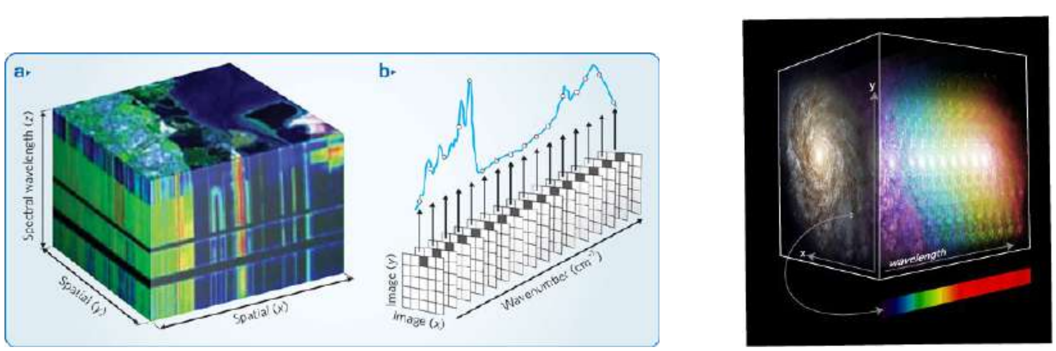


Figure 2.2: Illustrations of how a data cube looks like.

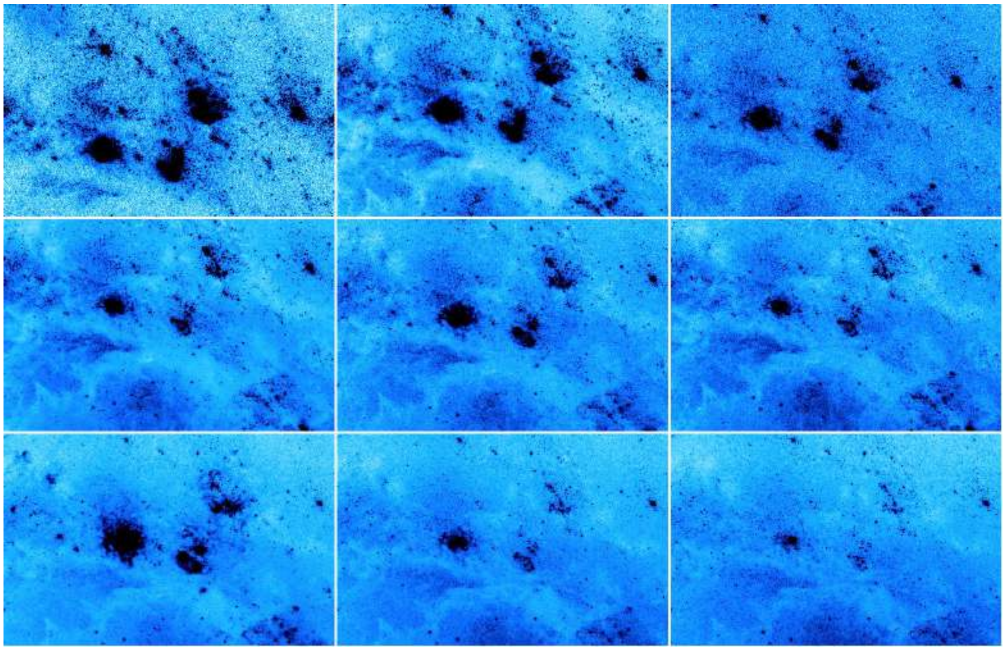


Figure 2.3: Example of how an object can look in 9 wavelengths

Take the time to think about this, how the data looks like in 3D, how a star looks like in the data cube, imagine it, this is where ideas of how to tackle this problem come from.

2.1.1 Topics you should review

This will require a lot of work, but hey it will be worthy and fun!

- Astroinformatics and computer science
  - Data mining
  - Machine Learning
  - Big Data Analysis
  - Neural Networks
  - Visualization Resources
- Statistics and Image Processing
  - Probability Density Function
  - Point Spread Function
  - Full width at half maximum
  - Convolution
- Interstellar medium and star formation
  - HII regions
  - Planetary Nebulae
  - Supernova Remnants
  - Molecular Gas
  - All kinds of Nebulae (e.g. dark, reflection)
  - AGN's (Active Galactic Nucleus)
- Astrophysics

- Units (light-years, parsecs)
- World coordinate system
- Light
- Telescopes
- Stars and Stellar Evolution
- Distance, Brightness, Luminosity
- Galaxies

The GitHub page will certainly help you to understand why you need to learn about that, and where to find articles, web pages and books.

2.1.2 Downloading

First, let's equip ourselves with the basic software you will need in order to start then you may probably find other cool programs and later you will install them. There is also the possibility that your assigned computer will have them installed already but here is a brief description of what you can do with them, most of them are easy to use.

DS9: It is a program that visualizes astronomy images in FITS format (don't worry if you recognize this format, it will be explained later), where you can easily manipulate them, read their headers, compare, look at regions, see their characteristics, make graphs, even videos. Well, depending on what you need to use later you will be finding all the functions, the best way is to click everywhere and find out what happens, also you can ask to your astronomy colleagues they will tell you all the perks, or if you like learning by yourself or you need something specific check the documentation web page. It is fairly easy to install, just follow the instructions.
Download: `http://ds9.si.edu/site/Download.html`
Documentation: `http://ds9.si.edu/site/Documentation.html`
The picture below shows (Fig.2.4something cool you can do in DS9.

Python and a user interface: The most *limitless* and user friendly way to develop programs in Astronomy is using Python, there are many packages, modules, functions now available to help you in almost anything. Me, as an undergrad engineer I'm used to program on an user interface and not directly in a terminal. So, here I will explain you my own way of doing things.
I make my programs on the Canopy editor, it shows when and where you have programming error and warnings, and the interface is easy to learn, now to run, I open a terminal, go to the directory where my program is, type `ipython` wait and then type `run myProgram.py`, and wait for the result.
Now there are a lot of fancier ways to work with *Python*, you can program and test directly using *IPython Notebook* on a web browser or you can just go for the terminal, use *nano* or *vi* or the text editor you like and then run it by typing `python myProgram.py`. At this point is up to you, but hey here are some links to start and the packages/modules you should install.
Interfaces or Development environments

- PyCharm, it a development environment, just like CodeBlocks or NetBeans `http://www.jetbrains.com/pycharm/`
- Spyder, actually this is the interface that comes with the Python distribution Anaconda, you will get the Python distribution and the interface. `https://store.continuum.io/cshop/anaconda/`

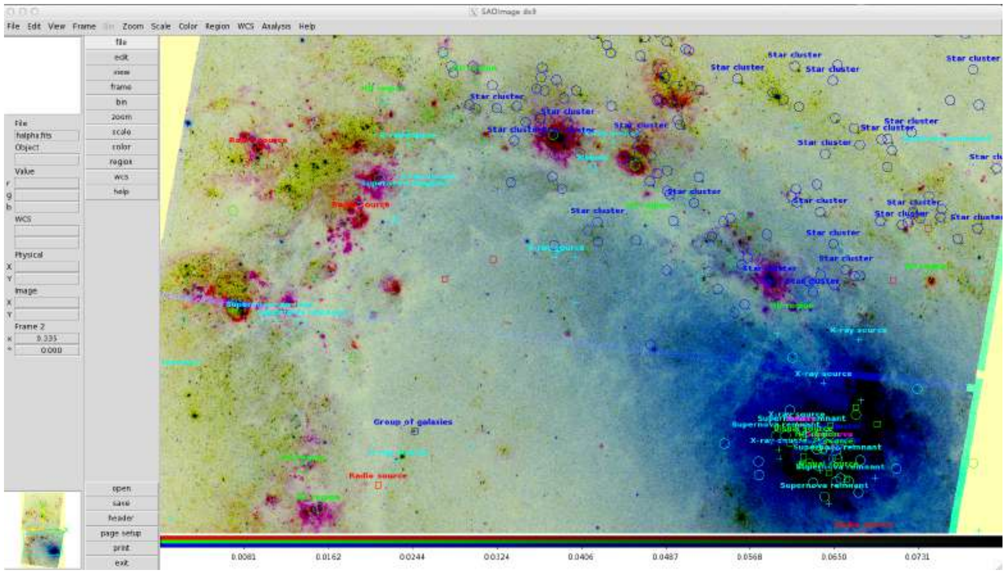


Figure 2.4: This is an RGB picture made from 3 independent FITS files, with a z scale and a region file overlaid from NED database, if you would like to learn more about this, or reproduce it, it is all explained in this web page: `https://github.com/LaurethTeX/Clustering/blob/master/NEDtoREGION-FILE/KnownRegions.md`

- Canopy, this is the one I mentioned before, it super easy to use and you can install packages with one click. `https://www.enthought.com/products/canopy/`

Modules

In Python, modules are like the libraries in C, therefore, to use math, astronomy and computer science tools you need to install them. To learn whether you already have a module installed or not, type on *iPython* `import andreaModule`, if the output result is something like `ImportError: No module named andreaModule`, you definitely don't have it installed.

The strategy here to install packages it fairly easy, find their website, go to the download section and follow the instructions, almost all the packages are available on the Python Packaging Index and may be installed by running:

```
pip install pyfits
```

To learn how to use them check the documentation page, user manuals or their API's, if you have experience on object oriented programming it will be like running a new bike and if you don't, don't worry too much, Python was designed to be easy to program, just learn the rules of the game.

- Astropy, this package is the *must have* of every astronomer, contains tools to handle coordinate systems, units, convolution.. well is better if you take a look at the web page. `http://www.astropy.org/`
- Numpy, this package contains the math magic functions, linear algebra tools and the array management variables, make sure you learn all about *Numpy arrays* you will work with them all the time. `http://www.numpy.org/`

- SciPy, well this package is the base of all scikit modules which contain the functions you will use in image processing and machine learning. `http://www.scipy.org/`
  - Scikit Image, contains image processing tools, it is the *OpenCV* for *Python* `http://scikit-image.org/`
  - Scikit Learn, contains data mining algorithms, pretty much contains everything that you will ever need. `http://scikit-learn.org/`
- Matplotlib, this package is probably one of the most powerful tools visualize data, you can draw almost anything you want and exactly how you want it. An example of that are the images of the AstroML book, you can access to the image library code and learn how they are made, this is the website `http://www.astroml.org/book_figures/index.html`.[2]. You can download the package here `http://matplotlib.org/`.
- PyFITS, in this package you will find tools to manipulate FITS files, create new ones, create image cubes, tables, and do all kinds of things with their headers. Certainly this package is more than useful. `http://www.stsci.edu/institute/software_hardware/pyfits`

In the path of researching I'm certain you will find more and new packages and by them you will be prepared to install anything.

Montage: This is a toolkit for assembling astronomical images into mosaics, but it has more functions that you may need in the future to prepare your data before processing it. There are two ways of installing and I would say that is better to have them both. One is to install the toolkit and any time you need it, you run the commands on the terminal, the other one is to install a *Python* module and use it just like any other module. To install montage for terminal, download the latest version in this website `http://montage.ipac.caltech.edu/docs/download.html`, **read the README file** or go to this website `http://montage.ipac.caltech.edu/docs/build.html` and follow the steps, now if you don't have any problem installing it, you can try testing it with an example program found on this website `http://montage.ipac.caltech.edu/docs/pleiades_tutorial.html`, in case you are having trouble and your computer is a MAC, instead of doing step five (*If you want to be able to run the Montage executables from any directory*), try this:

1. Open a file called `.profile` located in your user folder. (e.g. `/Users/Laureth`)

```
$ vi .profile
```

2. Include in the file the following

```
export PATH=/Applications/Montage_v3.3/bin:$PATH
```

   In this link (`https://github.com/LaurethTeX/Clustering/blob/master/Tools.md#the-profile-file`) you will find an example of how this file should look. After you modify it, make sure that you save it and type in `/Users/Laureth`,

```
$ source .profile
```

   Then try testing the *Montage* commands, and I'm sure that it will magically work, just

[2] Statistics, Data Mining, and Machine Learning in Astronomy book, it was mentioned before

remember that any time you use any command, type `source .profile`.

Now the other way to install, implies only to install a *Python* module but this module contains less functions that the terminal application, in any case check the website `http://www.astropy.org/montage-wrapper/`, there you will find all the documentation you may need and the instructions to install it (*Spoilers* `pip install montage-wrapper`).

Any questions you may have and how to install, here is my GitHub page for software tools `https://github.com/LaurethTeX/Clustering/blob/master/Tools.md`

3. Missions to the moons

As we know, there have been many missions to our Moon, some of them aimed at reaching the Moon, some landing on the Moon, and even returning samples to Earth. The data on the internet suggests there have been around 140 missions to the Moon to this date.
But out of all of them, some of the most interesting missions have been held in this decade, and yes, we will talk about them here.

3.1 The Artemis program

The Artemis program was led by NASA and initiated in 2017. It is aimed at returning humans to the Moon and establishing a sustainable presence there. It represents the first crewed lunar exploration effort since the Apollo program's final mission in 1972. The long-term goal of Artemis is to prepare for human missions to Mars and deeper space exploration. The Artemis program's infrastructure combines cutting-edge technology, multinational cooperation, and commercial partnerships to establish sustainable human exploration on the Moon and beyond.

3.2 Missions:

- Artemis I (2022): An uncrewed mission testing the Orion spacecraft and SLS rocket.
- Artemis II (planned for 2026): The first crewed mission to fly around the Moon.
- Artemis III (planned for 2027): The program's first crewed lunar landing near the Moon's south pole

3.3 Components:

Orion Spacecraft

The Orion spacecraft is the centrepiece of the Artemis missions. Designed for deep-space exploration, it provides crew transport to lunar orbit and back to Earth. Orion includes a pressurized crew module, advanced life support systems, and the European Service Module (ESM) developed by the European

Space Agency ESM supplies power, propulsion, and life support resources like water and oxygen. This spacecraft is engineered to withstand the intense conditions of deep space, offering unparalleled safety and reliability for astronauts.

3.3.1 Space Launch System (SLS)

The Space Launch System (SLS) is NASA's most powerful rocket, built to propel Orion and heavy payloads into space. The SLS is modular, allowing configurations for different missions.

3.3.2 Lunar Gateway

The Lunar Gateway is a modular space station that will orbit the Moon, supporting long-term lunar operations and serving as a staging point for crewed and robotic missions. The Gateway will house scientific laboratories, communication systems, and living quarters for astronauts.

3.3.3 Artemis Base Camp

On the Moon's surface, the Artemis Base Camp will serve as a long-term habitat and operations hub near the lunar south pole. The base will include facilities for living, conducting experiments, and resource utilization, such as extracting water ice for oxygen and fuel.

4. Tidal Forces and it's effects

Tidal forces arise because the gravitational pull from one planet (or moon) is not uniform across the other body. For example, the side of a planet closer to the other body experiences a slightly stronger gravitational pull than the far side. The strength of the gravitational pull that a planet exerts on another depends on the distance between them and their masses. The closer the bodies are and the more massive they are, the stronger the gravitational attraction.

4.1 Tidal bulges

Tidal bulges are the result of the gravitational interactions between the Earth, the Moon, and the Sun. These interactions create the rhythmic rise and fall of sea levels known as tides.
The Moon's gravitational force affects the Earth, but since water in the oceans is fluid, it is more responsive to these forces than the solid parts of Earth.On the side of Earth closest to the Moon, the Moon's gravity pulls water toward it, creating a bulge. This bulge is the high tide on the side facing the Moon.As water is redistributed to form the tidal bulges, there is less water in the areas perpendicular to the tidal bulge. These areas experience low tides because water is essentially "pulled away" to feed the bulges.
When the Moon first formed, it likely rotated much faster than it does today. The Earth's gravity exerted tidal forces on the Moon, creating tidal bulges on its surface The gravitational pull between Earth and the misaligned bulges created a torque that acted to slow down the Moon's rotation.Over millions of years, the Moon's rotation slowed to the point where its rotational period matched its orbital period of 27.3 days.At this point, the tidal bulges became aligned with the Earth-Moon axis, eliminating the torque that had been slowing the Moon's rotation.

4.2 Tidal locking

Tidal locking is a phenomenon that occurs when a moon or a smaller body's rotation period becomes synchronized with its orbital period. In this state, the moon or body always shows the same face to

the object it is orbiting, meaning that one hemisphere is constantly facing the planet, while the other hemisphere remains hidden.
While the Moon is tidally locked to Earth, Earth's rotation is also gradually slowing due to the Moon's tidal forces, although the process is much slower for Earth. This effect is causing the Moon to slowly move away from Earth by about 3.8 cm per year.

4.3 Orbital resonance

It refers to a phenomenon that occurs when two or more orbiting bodies, such as moons, planets, or asteroids, exert regular, periodic gravitational influences on each other due to their orbital periods being related in a simple ratio. For example, if two moons of a planet have a 2:1 resonance, for every two orbits one moon completes, the other moon completes one orbit. One of the most famous examples of orbital resonance involves Jupiter's moons Io, Europa, and Ganymede. These three moons are in a 1:2:4 orbital resonance. : In some cases, the gravitational interactions in a resonance can cause one or more of the bodies' orbits to become more eccentric (elongated) or more circular. This can lead to phenomena like tidal heating (in the case of moons) or changes in the shape of the orbit.

4.4 Tidal heating

– it is the process by which the gravitational interactions between two orbiting bodies (such as a planet and its moon, or a planet and its star) cause internal friction within one or both of the bodies, leading to the generation of heat. This heat can result from the deformation of the body due to tidal forces and can have significant geological and atmospheric effects on the affected body.

4.4.1 Effects of Tidal Heating

Tidal heating can cause significant geological activity, such as volcanism or tectonic movements, within the body experiencing the tidal forces. A prime example of this is Io, one of Jupiter's moons, which has intense volcanic activity driven by tidal heating. As Io's orbit is elliptical and is in resonance with other Galilean moons of Jupiter (Europa and Ganymede), the constant flexing of Io generates immense internal heat, causing frequent volcanic eruptions on its surface.
It can also have important implications for the habitability of moons or planets. For example, tidal heating is believed to keep the subsurface ocean of Europa, another moon of Jupiter, in a liquid state beneath its icy crust. The heat generated by tidal forces helps prevent the ocean from freezing solid, making Europa one of the most intriguing places to search for potential life in our solar system.

4.4.2 Tidal Heating: Mathematical Representation

Tidal heating depends on a body's orbital eccentricity, size, and composition.
The power dissipated as heat due to tidal deformation:
$P = k * e^2 * R^5 / p * Q$
e=Orbital eccentricity, p =Rigidity of the moon's material, Q=Dissipation factor, k= constant

4.5 Effects on Europa

Tidal heating is thought to be a major source of internal energy for Europa, and it is likely responsible for maintaining its subsurface ocean in a liquid state, despite the extremely cold temperatures on the surface. Without this internal heating, the ocean could freeze solid. It also contributes to the geological activity observed on Europa, such as ice tectonics, the movement and breaking of the icy surface. This is evident in the cracked and ridged terrain seen on Europa's surface.
It is hypothesized that beneath Europa's ice shell, which is estimated to be about 10–30 kilometers thick, there exists a liquid water ocean that could be in contact with the moon's rocky mantle. The heat generated by tidal forces could keep this ocean liquid, potentially creating a habitable environment for life.

4.6 Effects on Enceladus

One of the most remarkable features of Enceladus is its cryovolcanic activity, where geysers of water vapor, ice, and other volatile compounds shoot out from cracks in its icy surface, especially near the south pole. These geysers are powered by the tidal heating that creates the necessary pressure to eject material from the moon's interior. The material ejected by the geysers is believed to come from a subsurface ocean beneath Enceladus' icy crust. The heat generated by tidal forces may allow liquid water to exist beneath the icy shell, which is why the moon has been a target in the search for potential extraterrestrial life. The geysers eject water vapor, organic compounds, and salts, indicating that there may be habitable conditions in the subsurface ocean. These geysers were first discovered by the Cassini spacecraft in 2005.

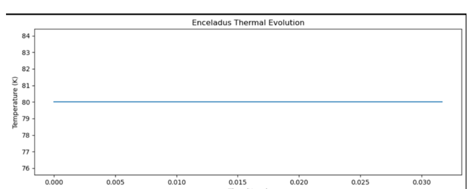
4.7 Tidal heating simulation

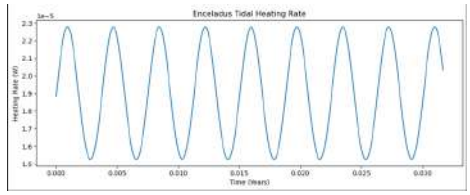
```python
def tidal_heating(M, n, R, a, e, k):
    """
    Calculate tidal heating rate

    Parameters:
    M - Mass of central body (kg)
    n - Mean motion (rad/s)
    R - Radius of moon (m)
    a - Semi-major axis (m)
    e - Orbital eccentricity
    k - Love number

    Returns:
    Tidal heating rate (W)
    """
    return (21/2) * M * (n)**3 * (R/a)**5 * k * e**2
```

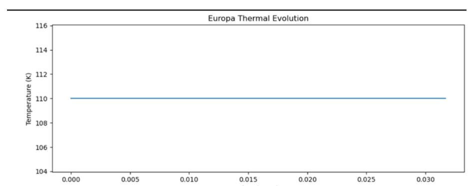
```python
class TidalHeatingSimulation:
    def __init__(self, moon_name):
        # Store moon name
        self.moon_name = moon_name

        # Moon-specific parameters
        if moon_name.lower() == 'enceladus':
            self.M_central = 5.683e26  # Mass of Saturn (kg)
            self.R_moon = 252.1e3  # Radius of Enceladus (m)
            self.a = 238.02e6  # Semi-major axis (m)
            self.e = 0.0047  # Orbital eccentricity
            self.k = 0.72  # Love number
            self.n = 5.3e-5  # Mean motion (rad/s)
            self.initial_temp = 80  # K
            self.heat_capacity = 2000  # J/kg·K (for icy material)
            self.mass = 1.08e20  # kg

        elif moon_name.lower() == 'europa':
            self.M_central = 1.898e27  # Mass of Jupiter (kg)
            self.R_moon = 1560.8e3  # Radius of Europa (m)
            self.a = 671.1e6  # Semi-major axis (m)
            self.e = 0.009  # Orbital eccentricity
            self.k = 0.37  # Love number
            self.n = 2.0548e-5  # Mean motion (rad/s)
            self.initial_temp = 110  # K
            self.heat_capacity = 2100  # J/kg·K (for icy material)
            self.mass = 4.80e22  # kg

        else:
            raise ValueError("Choose either 'Enceladus' or 'Europa'")

        self.G = 6.67430e-11  # Gravitational constant

    def tidal_heating_rate(self, t):
        """
```

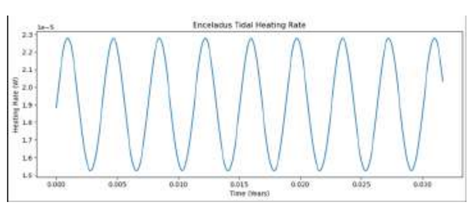
```python
        Calculate instantaneous tidal heating rate
        Accounts for orbital variations
        """
        # Basic tidal heating formula with time-dependent eccentricity
        time_varying_e = self.e * (1 + 0.1 * np.sin(self.n * t))
        return (21/2) * self.M_central * self.n**3 * (self.R_moon/self.a)**5 * self.k * time_varying_e**2

    def thermal_evolution(self, initial_temp, time_array):
        """
        Simulate thermal evolution due to tidal heating
        """
        def temp_change(T, t):
            # Calculate heat input
            heating_rate = self.tidal_heating_rate(t)

            # Temperature change rate
            dTdt = heating_rate / (self.mass * self.heat_capacity)
            return dTdt

        # Solve differential equation
        temps = odeint(temp_change, initial_temp, time_array)
        return temps.flatten()

    def simulate(self, duration=1e6, time_steps=1000):
        """
        Run full simulation
        """
        # Time array (seconds)
        time_array = np.linspace(0, duration, time_steps)

        # Simulate heating
        temperatures = self.thermal_evolution(self.initial_temp, time_array)
        heating_rates = [self.tidal_heating_rate(t) for t in time_array]
```

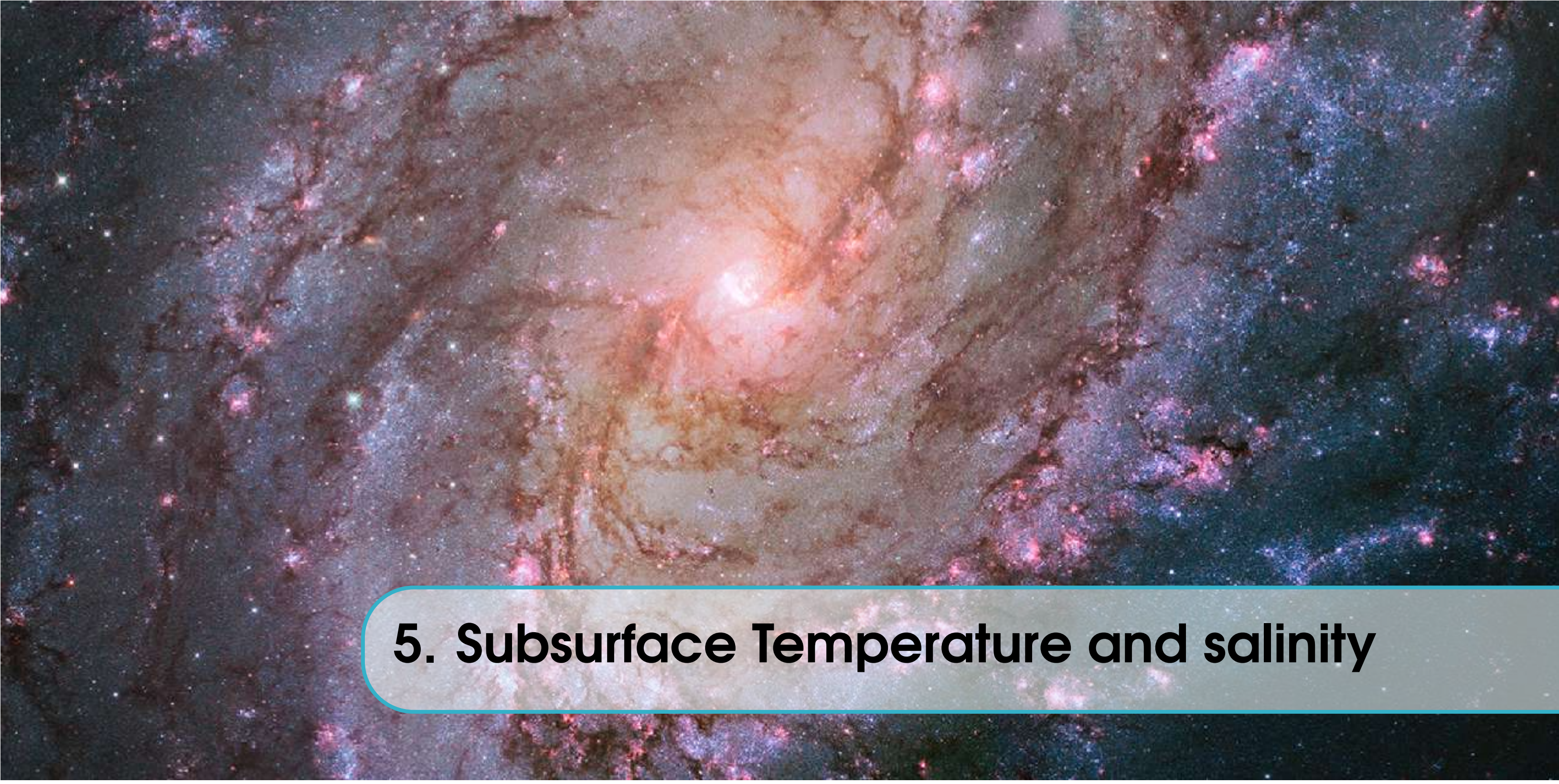
```python
        # Plotting
        fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 8))

        # Temperature evolution
        ax1.plot(time_array / (365.25 * 24 * 3600), temperatures)
        ax1.set_title(f'{self.moon_name.capitalize()} Thermal Evolution')
        ax1.set_xlabel('Time (Years)')
        ax1.set_ylabel('Temperature (K)')

        # Heating rate evolution
        ax2.plot(time_array / (365.25 * 24 * 3600), heating_rates)
        ax2.set_title(f'{self.moon_name.capitalize()} Tidal Heating Rate')
        ax2.set_xlabel('Time (Years)')
        ax2.set_ylabel('Heating Rate (W)')

        plt.tight_layout()
        plt.show()

        return {
            'time': time_array,
            'temperatures': temperatures,
            'heating_rates': heating_rates
        }
```




Enceladus Tidal Heating Rate
Heating Rate (W)
Time (Years)


Europa Thermal Evolution
Temperature (K)
Time (Years)


Enceladus Tidal Heating Rate
Heating Rate (W)
Time (Years)



5. Subsurface Temperature and salinity

Introduction

The subsurface temperature of icy moons like Europa and Enceladus plays a crucial role in understanding their geology, overall composition, and, most importantly, their potential for harboring life.

Subsurface Temperature

Subsurface temperature refers to the temperature beneath the outermost surface of an icy moon. It can vary significantly depending on depth, internal heat sources of the moon, and its distance from external heat sources like the Sun.

Temperature Variation

The variation of subsurface temperature is expected to occur in the following manner:

1. Near the Icy Surface (Outer Layer)

The temperature just below the exposed icy layer is very low, but due to insulation caused by the outer icy layers, typically composed of frozen water and other ices (like CO_2 or methane), the temperature is somewhat higher compared to the outermost surface.

Example: On Europa, the near-subsurface temperature might range from 110–130 K, while on Enceladus, it could be 80–100 K near the poles.

2. Cryosphere

Temperature rises due to heat conduction from the moon's interior and reduced heat loss to space as the depth within the ice shell increases. Tidal forces and radioactive decay may further elevate temperature at the base of the ice.

Example: The temperature at the bottom of Europa's 20–30 km thick ice shell could be around 270 K, enabling the ice to transition to liquid water.

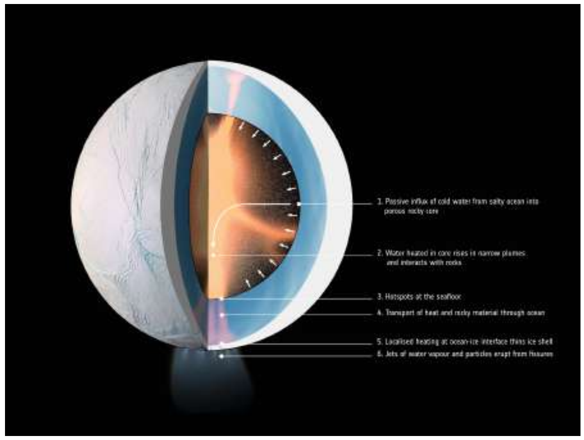


Figure 5.1: Geotherm

3. Subsurface Oceans

The temperature in these regions of the moons with subsurface oceans is maintained at or above the freezing point of water (273 K) depending on pressure and salinity.

Ammonia or other antifreeze compounds can lower the freezing point, maintaining liquid water at lower temperatures (e.g., 176–273 K for Titan).

Example: Europa's ocean may hover around 273 K, while Titan's ocean, rich in ammonia, could be closer to 180 K.

4. Rocky Core

The highest subsurface temperature is found in rocky cores, heated by radioactive decay and tidal friction in some cases. These temperatures can exceed 400 K near the core-ocean interface, particularly if hydrothermal activity is present (as hypothesized for Enceladus).

Factors Affecting Subsurface Temperature

Sources of Heat

- **Radioactive Decay:** Heating caused by the decay of isotopic substances like Uranium, Thorium, and Potassium in the rocky cores.
- **Tidal Heating:** Creation of internal friction due to gravitational interactions of icy moons like Europa and Enceladus with their parent planets or other moons.
- **Residual Heat:** Heat left after the formation of the moon, especially in older moons.

Thermal Layers (Geotherm)

Thermal layers refer to the temperature regions from the frozen surface to the warm interior. These layers help understand heat distribution and convection through the moon's crust.

Geotherm Overview

- **Upper Surface:** Temperatures on the surface are extremely cold due to the distance from the Sun (e.g., Europa: 110 K, Enceladus: 75 K).
- **Ice Shell:** Below the surface, the temperature rises due to the insulating properties of ice and internal heating.

- **Subsurface Ocean:** For moons with oceans beneath the ice (e.g., Europa, Ganymede, Enceladus), the temperature can be above freezing, potentially supporting liquid water.
- **Core:** The core is the hottest region, with temperatures depending on the moon's composition and internal heat generation.

Moons with Subsurface Potential

- **Europa (Jupiter):** Expected to have a subsurface ocean beneath a 10–30 km thick ice crust. Tidal heating maintains the ocean in liquid form.
- **Enceladus (Saturn):** Geysers at the south pole suggest a subsurface ocean beneath -5 km of ice. Tidal heating plays a significant role.
- **Ganymede (Jupiter):** Magnetic field is indicative of a subsurface salty ocean beneath a thicker ice crust.
- **Titan (Saturn):** Thick icy shell over a possible subsurface ocean; its unique atmosphere and chemistry make it intriguing.

6. Research Insights into Icy Ocean Worlds

6.1 1.SEA ICE, EXTREMOPHILES AND LIFE ON EXTRA-TERRESTRIAL OCEAN WORLDS

In a research carried out by Andrew Martin and Andrew McMinn on the world of sea-ice microbes and their potential role in the search for life on ocean worlds like Europa and Enceladus. Despite having harsh and unfriendly conditions, there could be some warmth welcoming corners with ice, the ocean beneath, and the ocean floor.

This research was carried out by examining the adaptation of extremophiles, particularly sea-ice microorganisms on Earth, as models for understanding life's adaptability and resilience in extreme environments. Some of the challenges faced in figuring out the potential energy sources could come from, as well as understanding how life thrives and how the ecosystems supporting life are created in such extreme conditions. The tiny extremophiles which thrive in extreme environments on Earth are the perfect candidates for studying potential life forms on other planets.

Microorganisms are vital for Earth's health, playing a key role in energy transfer and chemical reactions in oceans. They outnumber stars in the universe and have been around for billions of years, showcasing incredible diversity and adaptability. Their genetic potential holds promise for medical and biotechnological advancements. Despite our knowledge, much remains to be discovered about these resilient organisms.

They have drawn intuition that these microbes might be a universal foundation for life in the universe. This means if life thrives on celestial bodies, it could resemble what is there on Earth. It is still a mystery how life began on Earth, but studying how these organisms are able to survive such harsh conditions provides us with clues about the potential for life elsewhere in the universe.

A fascinating approach was used by the authors in defining the term *HABITABILITY* in terms of supporting at least one known organism. It simplifies the concept and makes it more practical for scientific exploration. This definition, proposed by Cockell et al. (2016), allows researchers to focus on identifying environments that can sustain life as we know it, without getting bogged down in the complexities of defining life itself. It also helps in understanding the potential longevity and adaptability of organisms in various habitats. Additionally, this concept connects closely to how far habitability extends, giving us context about the organisms that might be present and how long they can thrive.

6.1.1 SUMMARY

In short, the insights we gather from Earth's extremophiles can be incredibly valuable as we seek to understand the possibilities for life in the mysterious environments beyond our planet.

6.2 2. HOW DOES SALINITY SHAPE OCEAN CIRCULATION AND ICE GEOMETRY ON ENCELADUS AND OTHER ICY SATELLITES?

The recent research by Wanying Kang and her team examines how salinity affects ocean circulation and ice structure on Enceladus. This moon is of great interest to scientists because it features a global saline ocean underneath a thick ice shell, hinting at interactions between water and rock that could make it a candidate for habitability.

6.2.1 Key Highlights from the Study:

1. **Ocean Dynamics and Salinity :** The researchers discovered that salinity influences ocean circulation greatly. At extreme salinity levels, circulation patterns behave differently, with fresh water and heat tending to flow toward the equator, where the ice is thicker.
2. **Heat Sources and Their Impact :** Potential heat sources like hydrothermal vents and tidal heating could keep the ocean in a liquid state. However, there's still uncertainty about which source plays the biggest role.
3. **Salinity Levels :** The salinity of Enceladus' ocean is estimated to range from 2 to 20 practical salinity units (PSU), but it might often be lower than 10 PSU. The Cassini spacecraft's samples suggest salinity levels between 5 and 20 PSU.
4. **Exciting Astrobiological Potential :** The geyser samples contain chemicals like CO2, methane, sodium salts, and organic materials, suggesting a lively chemical environment beneath the ice that could be conducive to life.
5. **Modeling Challenges:** Accurately modeling heat movement in this complex system is challenging. Current models struggle to generate enough heat to keep the ice shell as thin as observed.

6.2.2 SUMMARY:

This study reveals the fascinating and complex interactions between salinity, ocean dynamics, and ice structures on Enceladus. It helps paint a clearer picture of the moon's potential for supporting life and sets the stage for future discoveries as we continue to explore what lies beneath its icy surface.

6.3 3.DYNAMIC EUROPA OCEAN SHOWS TRANSIENT TAYLOR COLUMNS AND CONVECTION DRIVEN BY ICE MELTING AND SALINITY

Yosef Ashkenazy and Eli Tziperman explored the ocean beneath Europa's thick ice to understand its potential for supporting extraterrestrial life. They used a high-resolution General Circulation Model (GCM) to study how heating, melting, and freezing ice affect salinity. Here are the main findings:

- **Salinity's Importance**: The ocean's weak stratification is influenced by salinity variations, affecting heat and mass movement.

- **Active Convection**: The ocean shows strong transient convection patterns, creating swirling eddies and zonal jets.
- **Taylor Columns**: Ocean currents form Taylor columns, aligning with Europa's rotation axis and moving toward the equator outside stable areas.
- **Heat Transport**: Intense heat transport leads to a nearly uniform ice thickness above the ocean.
- **Better Modeling Techniques**: The improved methods allow the model to determine the dynamics of the ocean temperature with greater precision.

6.3.1 SUMMARY

This research highlights the importance of considering salinity and non-hydrostatic effects to better understand Europa's ocean and its potential for life.

6.4 4.ENCELADUS'S OCEANS MAY HAVE THE RIGHT SALTINESS TO SUSTAIN LIFE

This article discusses new insights into the potential habitability of Saturn's moon Enceladus, focusing on the salinity of its subsurface ocean and the factors that could make it conducive to life. Here's a more detailed breakdown of the findings and their implications:

6.4.1 1. Surface Ice and Ocean Salinity

Enceladus has a surface completely covered by clean, bright ice, which is a major feature of its appearance. Understanding the characteristics of this ice shell is key to determining the properties of the ocean beneath it, as the ice acts as a barrier to studying the liquid water directly.

Researches, led by Wanying Kang from MIT, wanted to determine the salinity of the subsurface ocean beneath Enceladus's ice shell. The findings suggest that the ocean's salinity is close to Earth's, with the salt concentration potentially around 30 grams of salt per kilogram of water. Earth's oceans, by comparison, have a salinity of about 35 grams of salt per kilogram. This is a significant finding because the salinity of the ocean is an important factor in determining its ability to support life.

6.4.2 2. Theoretical Model

To study the relationship between ocean salinity, ice thickness, and the properties of the water beneath the ice, the research team developed a theoretical model. This model took into account how ocean currents, salinity, and ice geometry influence each other. By adjusting the model to replicate the specific features of Enceladus's ice, they were able to infer details about the subsurface ocean.

One of the key observations made by the team is that on Enceladus, the ice is thinner at the poles than at the equator. This variation in ice thickness corresponds to the salinity of the ocean. Thicker ice at the equator and thinner ice at the poles suggest that the ocean's salinity is relatively high, making the conditions somewhat similar to Earth's oceans.

6.4.3 3. Hydrothermal Vents and Heat

- Another crucial aspect of the study was the discovery of heat emanating from the bottom of Enceladus's ocean. This could indicate the presence of hydrothermal vents, similar to those found on Earth. On our planet, hydrothermal vents are known to support life, as they provide the necessary heat and chemical nutrients for organisms to thrive in otherwise extreme environments.

- The idea that hydrothermal vents might exist on Enceladus and could be a source of energy for potential microbial life strengthens the case for the moon's habitability. This is important because, on Earth, life forms often flourish around such vents, relying on the chemical energy produced by the vents rather than sunlight.

6.4.4 4. WATER CIRCULATION AND TEMPERATURE

- Understanding the circulation of water beneath the icy shell is another crucial factor in determining habitability. The researchers found that temperature differences in the water are likely driving currents, which are essential for nutrient mixing and maintaining stable conditions in the ocean.
- The presence of liquid water in circulation is important because it suggests that the ocean could have a dynamic environment, similar to Earth's oceans, which supports life by redistributing heat, minerals, and other essential elements.

6.4.5 5. IMPLICATION ON LIFE AND FUTURE RESEARCH

The study adds to the growing body of evidence that Enceladus could support life, particularly because of its relatively Earth-like conditions, including the presence of organic molecules (detected by the Cassini spacecraft in plumes of water vapor), the right salinity, and potential heat sources like hydrothermal vents.

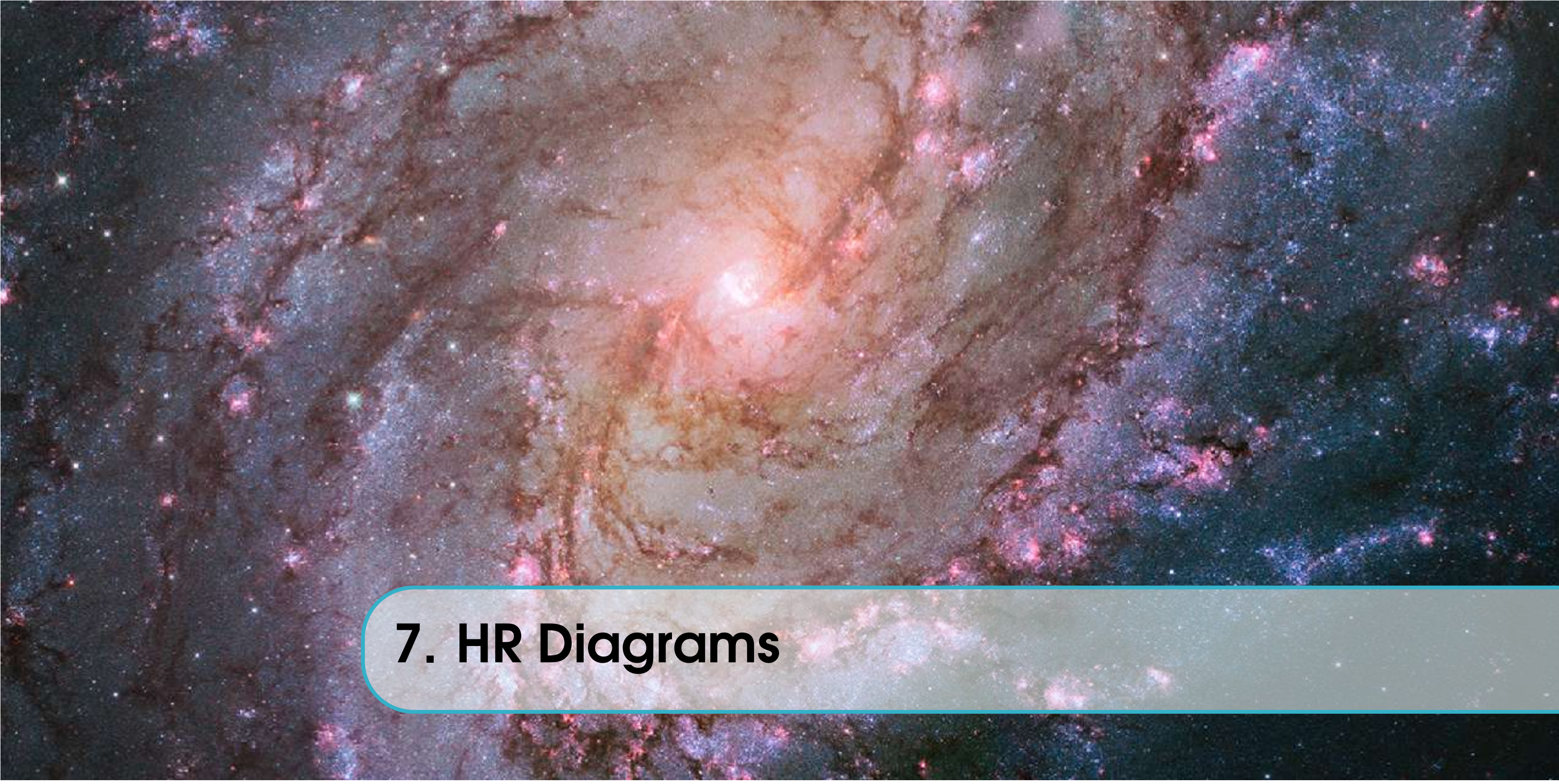
While salinity is just one of the factors affecting the habitability of an ocean, it is a significant one. The salinity of Enceladus's ocean suggests that, in terms of its capacity to support life, it might be quite similar to Earth's oceans. However, salinity alone isn't enough to guarantee life; temperature, chemical composition, and other factors play important roles.

6.4.6 6. Next Steps: Europa and Other Icy Moons

The team is now applying their model to Jupiter's moon Europa, which is believed to have a higher salinity than ocean of Enceladus. Europa, like Enceladus, is considered a promising candidate for harbouring life beneath its icy surface.

6.4.7 SUMMARY

The ultimate goal of this research is to refine models of the subsurface oceans of all icy moons and planets observed in our solar system. As missions continue to explore these moons, the insights gained from models like Kang's could help scientists better assess their habitability and potential for supporting life.



7. HR Diagrams

The HR Diagram, developed in the early 20th century by Ejnar Hertzsprung and Henry Norris Russell, classifies stars based on their temperature, luminosity, and spectral characteristics. Early 19th-century advancements in spectroscopy and parallax measurements laid the groundwork for these classifications. The HR Diagram plots stellar temperature (x-axis) versus luminosity (y-axis).

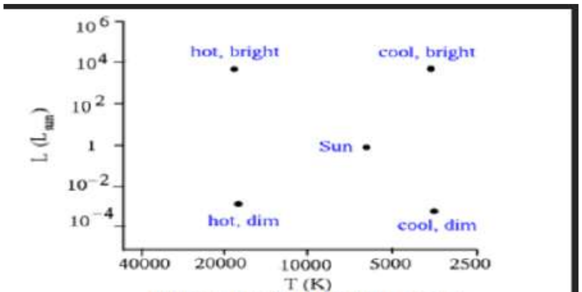


Figure 1. Basic HR Diagram Layout

• Temperatures are measured in Kelvin, with hotter stars on the left.
• Luminosity is relative to the Sun, with more luminous stars at the top.
• It categorises stars into spectral classes (O, B, A, F, G, K, M) based on decreasing temperature, with additional classes (L, T) for cooler stars.

7.1 Key Regions in the HR Diagram

7.1.1 Main Sequence

The main sequence is the dominant diagonal band on the HR Diagram, where stars spend most of their lifetimes burning hydrogen into helium in their cores. Stars in this region, including the Sun, range from massive, hot, blue stars at the upper left to cooler, less massive, red stars at the lower right.

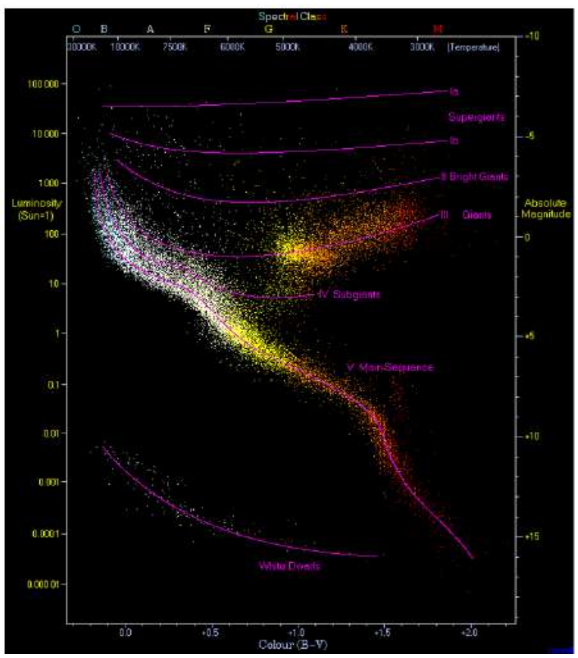
The Harvard Spectral Sequence			
Spectral Class	Distinguishing Characteristics	Temperature (K)	Example
O	Ionized He and metals; weak H	28,000 - 60,000	zeta Orionis
B	Neutral He, ionized metals, stronger H	10,000 - 28,000	Rigel, Spica
A	Balmer H dominant, singly-ionized metals	7500 - 10,000	Sirius, Deneb
F	H weaker, neutral and singly-ionized metals	6000 - 7500	Procyon, Canopus
G	Singly ionized Ca, H weaker, neutral metals	5000 - 6000	Sun, Capella
K	Neutral metals, molecular bands begin to appear	3500 - 5000	Aldebaran, Arcturus
M	Ti oxide molecular lines; neutral metals	< 3500	Antares, Betelgeuse

7.1.2 Giants and Supergiants

Giant and supergiant stars are bright and expansive in the upper right of the HR Diagram. They represent the late stages of stellar evolution when stars have exhausted their hydrogen fuel.

7.1.3 White Dwarfs

: Found in the lower left of the HR Diagram, white dwarfs are remnants of stars that have shed their outer layers after exhausting their fuel. They are hot but dim due to their small size. Though not conducive to habitability directly, planets or moons near white dwarfs might remain warm enough under specific conditions for subsurface habitability.



7.1.4 Habitability in icy ocean worlds

The study focuses on using geophysical measurements to investigate the internal structures and habitability of icy ocean worlds such as Europa and Enceladus. By examining these parameters, researchers aim to determine whether these environments might support life. Understanding the physical and chemical conditions of these celestial bodies helps refine our search for extra-terrestrial habitability within the solar system.

Key parameters, including density, temperature, sound speed, and electrical conductivity, are analyzed to model the internal environments of these icy moons. These factors reveal the state of subsurface oceans, their composition, and interactions with underlying rock layers. Such insights are crucial for assessing whether these oceans provide conditions conducive to life, such as liquid water, essential nutrients, and energy sources.

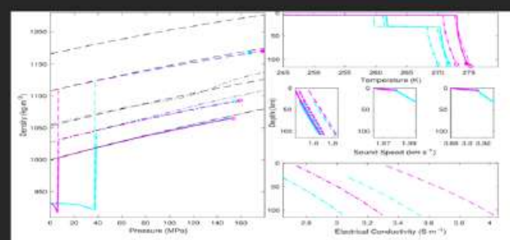


Figure 6. Europa: Oceans with 10 wt% $MgSO_4$(aq) (–), pure water (- -), and standard seawater (S_0=35.165 g per kg fluid; -·-) for T_b as in Table 6. Left: Density versus pressure. Reference fluid densities along the melting curve for $MgSO_4$ are {0,5,10,15} wt% and for seawater are {0,1,2}×S_0. Right: Corresponding depth-dependent temperature (top), sound speed in the fluids and ices (fluid, V_S, V_P middle left to right), and electrical conductivity (bottom). Circles indicate the transition to the silicate interior.

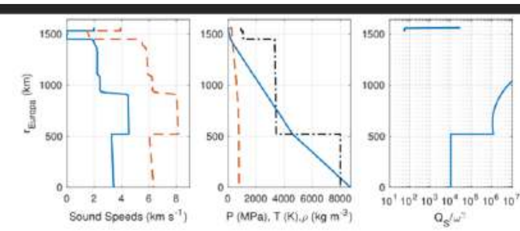


Figure 7. Europa: global interior structure for an ocean with the composition seawater and 30km-thick ice Ih lithosphere. Left: V_S (—) and V_P (- -). Middle: Pressure (—), temperature (- -), and density (-·-). Right: Anelasticity.

7.2 Europa: Trends and Insights

7.2.1 Graphical Trends

Temperature vs. Depth:

As depth increases, temperature rises steadily due to the compressibility of fluids and heat flux within the ocean layers. Oceans with higher salinity (e.g., MgSO-dominated) show a steeper temperature gradient compared to less saline oceans.

Density vs. Pressure:

Ocean density increases with pressure. Oceans with MgSO (10 wt percentage) exhibit a higher density profile than pure water or standard seawater. The differences highlight how salinity influences the compressibility and overall stability of the ocean.

Sound Speed (Vp and Vs):

Sound speeds in oceans increase with depth, reflecting higher pressures and the corresponding changes in material properties. Oceans with higher salinity (e.g., MgSO) exhibit significantly higher sound speeds compared to pure water due to ionic effects.

Electrical Conductivity:

Electrical conductivity increases with depth and temperature, with MgSO-dominated oceans having much higher values compared to seawater or pure water. This trend underlines the importance of ionic concentration

7.2.2 Theoretical Insights

Ice Thickness and Ocean Composition:

The ice thickness ranges from 5 to 30 km, with thinner regions more likely in geologically active zones. Higher salinity oceans suppress ice melting, affecting ice dynamics and surface interactions.

Habitability Implications:

The distinct geophysical signatures of chloride- vs. sulfate-dominated oceans provide clues about Europa's redox evolution and the chemical gradients that might support life.

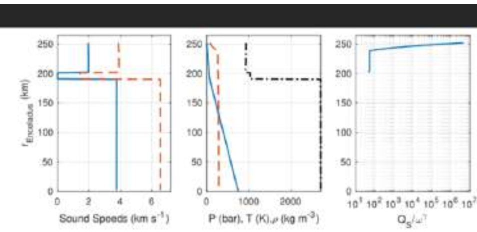


Figure 9. Enceladus: global interior structure with the ocean composition of seawater. Left: V_S (—) and V_P (– –). Middle: Pressure (—), temperature (– –), and density (· –). Right: Anelasticity.

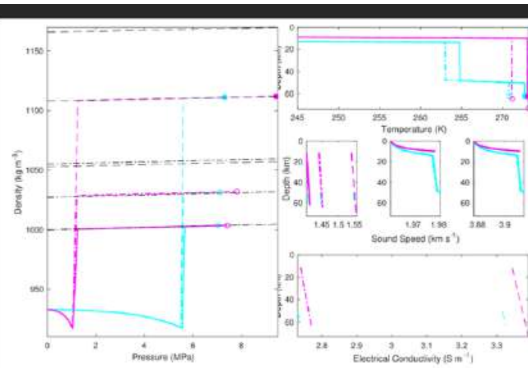


Figure 8. Enceladus: Oceans with 10 wt% $MgSO_4$(aq) (–), pure water (· ·), and standard seawater (– ·) for T_b as in Table 7. Left: Density versus pressure. Reference densities as for Europa (Fig. 6. Right: Corresponding depth-dependent temperature (top), sound speed in the fluids and ices (fluid, V_S, V_P middle left to right), and electrical conductivity (bottom). Circles indicate the transition to the silicate interior.

7.3 Enceladus: Trends and Insights

7.3.1 Graphical Trends

Temperature vs. Depth:

Temperature increases more sharply with depth in Enceladus compared to Europa, reflecting its smaller size and relatively higher thermal gradient. Salinity significantly affects the thermal profile;

MgSO oceans show steeper gradients compared to pure water.

Density vs. Pressure:
Density trends are analogous to Europa, with salinity exerting a strong influence. MgSO solutions are denser, leading to a distinct pressure-density curve.

Sound Speed (Vp and Vs):
Sound speeds increase with depth but are more pronounced in saline oceans due to higher compressibility. Variability in sound speeds across regions of Enceladus could reveal the presence of localized activity or compositional differences.

Electrical Conductivity:
Conductivity trends align with Europa, showing a marked increase with depth and temperature. MgSO solutions have much higher conductivity than pure water or seawater.

7.3.2 Theoretical Insights

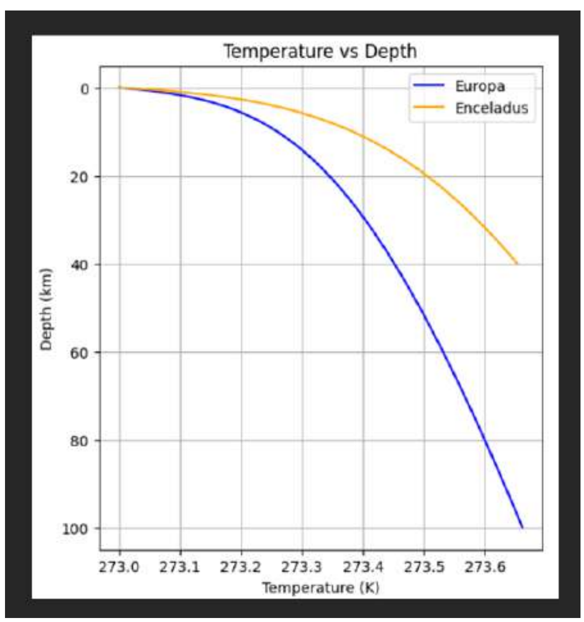
Ice Thickness and Ocean Composition:
Ice thickness varies significantly, from 10 km at the south pole to 50 km globally. Tidal heating plays a critical role, especially in active regions, maintaining a thin ice shell and potentially driving hydrothermal circulation at the ocean-floor interface.

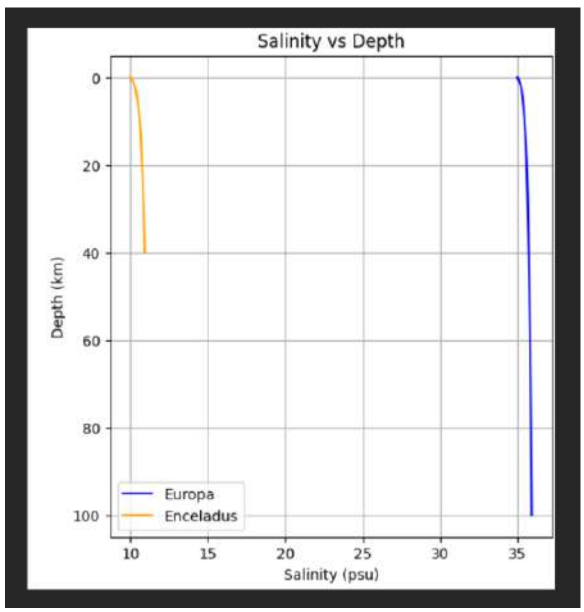
Rocky Core and Ocean Interaction:
The low-density rocky core may be either porous or hydrated. These characteristics affect the thermal and chemical fluxes between the core and ocean, influencing plume activity and habitability.

Habitability and Observational Potential:
The geophysical variations observed in density, conductivity, and temperature-depth profiles suggest that Enceladus's ocean is dynamically interacting with its core, supporting the production of hydrogen and other plume materials.

7.4 Comparative Analysis of Subsurface Ocean Properties: Europa vs. Enceladus

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters for Europa
depth_europa = np.linspace(0, 100, 1000)  # Depth in km
pressure_europa = depth_europa * 0.1  # Approximate pressure in MPa

# Adiabatic temperature gradient and MgSO4 salinity effect
temp_europa = 273 + 0.02 * pressure_europa + 0.1 * np.log(1 + depth_europa)
salinity_europa = 35 + 0.2 * np.log(1 + depth_europa)

# Parameters for Enceladus
depth_enceladus = np.linspace(0, 40, 1000)  # Depth in km
pressure_enceladus = depth_enceladus * 0.08

# Hydrothermal heating effect and high pH ocean dominated by NaCl
temp_enceladus = 273 + 0.03 * pressure_enceladus + 0.15 * np.log(1 + depth_enceladus)
salinity_enceladus = 10 + 0.25 * np.log(1 + depth_enceladus)

# Plot Temperature vs Depth
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.plot(temp_europa, depth_europa, label='Europa', color='blue')
plt.plot(temp_enceladus, depth_enceladus, label='Enceladus', color='orange')
plt.gca().invert_yaxis()
plt.xlabel('Temperature (K)')
plt.ylabel('Depth (km)')
plt.title('Temperature vs Depth')
plt.legend()
plt.grid(True)

plt.subplot(1, 2, 2)
plt.plot(salinity_europa, depth_europa, label='Europa', color='blue')
plt.plot(salinity_enceladus, depth_enceladus, label='Enceladus', color='orange')
plt.gca().invert_yaxis()
plt.xlabel('Salinity (psu)')
plt.ylabel('Depth (km)')
plt.title('Salinity vs Depth')
plt.legend()
plt.grid(True)
```




Salinity vs Depth
Depth (km)
Salinity (psu)
0
20
40
60
80
100
10
15
20
25
30
35
Europa
Enceladus